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On the relationship between saplings and ingrowth in northern hardwood stands

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ABSTRACT

We used long-term data collected from 22 study sites in northern hardwood stands comprised of sugar maple (*Acer saccharum* Marsh.), yellow birch (*Betula alleghaniensis* Britt.), and American beech (*Fagus grandifolia* Ehrh.) to establish relationships between sapling abundance and tree ingrowth. After 10 years, postharvest sapling density in the 6 cm diameter class (5.1–7.0 cm) showed linear relationships with ingrowth. Proportion of variation explained (r^2) varied from 36% to 83% depending upon tree species and silvicultural treatment (partial cutting vs. uncut control). After 20 years, linear relationships were also established (r^2 = 24–65%) between ingrowth and sapling density in the 2 cm diameter class (1.1–3.0 cm). From a wide pool of variables related to stand species composition, climate, physiography, and soil nutrients, postharvest sapling density was most strongly correlated to merchantable tree density (r = 0.43–0.75). Sugar maple sapling density was also positively correlated with base saturation and calcium saturation of the B horizon (r = 0.56 and 0.58). Over a 30-year period, the increase in American beech sapling basal area was substantial compared to mitigated increases found in sugar maple and yellow birch depending upon treatment. Our results provide useful information on integration of sapling data into forest management.

1. Introduction

In northeastern North America, hardwood forests provide a myriad of ecosystem services, including habitat for vertebrate wildlife species, recreational opportunities, watershed protection, and wood products (Leak et al., 2014). Deciduous angiosperms predominate in these forests, from shade-tolerant species like sugar maple (*Acer saccharum* Marsh.) and American beech (*Fagus grandifolia* Ehrh.) to mid-tolerant species like yellow birch (*Betula alleghaniensis* Britt.). Although each species has commercial value, sugar maple and yellow birch are generally more desirable than beech. Northern hardwood forests are often uneven-aged and characterized by small-scale disturbance that allows saplings (diameter at breast height (DBH) >1.0 and ⩽9.0 cm) to grow to merchantable size (>9.0 cm DBH). Historically, single-tree selection and group selection have been the main silvicultural treatments recommended to regenerate uneven-aged northern hardwood stands (Nyland, 1996; Leak et al., 2014). Although long-term responses of these partial cutting treatments on stand yield and diameter growth of individual trees are well understood (e.g. Bédard et al., 2012; Swift et al., 2012; Leak et al., 2014), there is less information on recruitment of saplings into merchantable size (Donoso et al., 2000). How many saplings are needed to obtain at least one recruit? How much time is required for a sapling to reach merchantable size? Does sapling size or species matter? Is there a minimum threshold of sapling abundance required to ensure the stand's long-term sustainability? What potential factors help explain sapling abundance, including silvicultural practices? These are important questions that have received less attention in the scientific literature compared to growth of merchantable trees and stand yield. Given that ingrowth is essential to ensure the long-term sustainability of a stand, there is a need to increase our knowledge to better manage northern hardwood forests.

Recent literature on long-term sapling dynamics in northern hardwood forests has documented an increase in beech over sugar maple over the past 40–50 years. Examples are found in uncut stands of several Canadian provinces and American States (Bedison et al., 2007; Duchesne and Ouimet, 2009; Gravel et al., 2011; Sullivan et al., 2013). Soil cation depletion appears to be an important limiting factor of sugar maple regeneration survival in forests affected by acid deposition (e.g., Moore et al., 2012; Marlow and Peart, 2014). Other studies, however, suggest that the increase in beech over maple cannot be explained solely by

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the soil cation depletion hypothesis (Gravel et al., 2010, 2011). Additional contributing factors may include preferential browsing by increasing white-tailed deer (*Odocoileus virginianus* Zimm.) populations on maple (Long et al., 2007; Matonis et al., 2011), and increased competitiveness of beech due to asexual reproduction (Jones and Raynal, 1986; Beaudet and Messier, 2008; Wagner et al., 2010). Beech bark disease (BBD) is another important issue affecting regeneration dynamics in northern hardwood forests across northeastern North America (Griffin et al., 2003; Morin et al., 2007). This disease is caused by feeding of the beech scale insect (*Cryptococcus fagisuga* Lind.) and subsequent infestation by fungi (e.g., *Nectria galligena* Bres. in Strass) that kills a large proportion of mature beech. Impacts of BBD on regeneration dynamics are still uncertain (Evans et al., 2005). Thus, we aim to build on this body of knowledge by investigating the influence of silvicultural practices and potential integration of sapling data into forest management. Studies that deal with this aspect in the literature are rare (Nolet et al., 2008; Bannon et al., 2015).

Our study uses up to 30 years of empirical data gathered from 22 study sites to investigate relationships between saplings and tree ingrowth (>9.0 cm DBH) in managed and unmanaged northern hardwood forests. Our data also offers an opportunity to examine changes in sapling abundance after harvest for three of the most common species: sugar maple, American beech, and yellow birch. Hence, for each species the objectives of this study were to: (1) establish relationships between sapling abundance, i.e., density or basal area (BA), and ingrowth after 10 and 20 years with or without partial cutting; (2) determine factors that help explain sapling abundance, such as overstory composition, climate, and soil properties; (3) quantify the change in sapling abundance over 30 years.

2. Materials and methods

2.1. Study area

Data used in this article were pooled from 22 study sites across the province of Québec, Canada (Fig. 1). Sites were located in the northern temperate vegetation zone (Saucier et al., 1998), and covered three bioclimatic domains across a latitudinal gradient (45.5–47.9°N): sugar maple – American basswood (*Tilia americana* L.), sugar maple – yellow birch, and balsam fir (*Abies balsamea* (L.) Mill.) – yellow birch. A longitudinal gradient (67.1–78.6°W) was also present, with a few sites extended in the western and eastern regions of Québec (Fig. 1). Depending upon treatment, mean postharvest density of merchantable trees ranged from 393 to 510 stems ha^{-1} and mean stand BA ranged from 18.8 to 26.2 m^2 ha^{-1} (Table 1). Species composition was generally dominated by sugar maple (>50%), followed by yellow birch and American beech (Table 1). Mean preharvest sapling density was 809 stems ha^{-1}, with sugar maple accounting for 61%, followed by American beech (34%), and yellow birch (4%). Postharvest sapling density was just below 560 stems ha^{-1}, harvesting operations reduced sapling density by 30% (Table 2).

2.2. Experimental design and silvicultural treatments

Partial cutting trials ($n = 65$) were carried out in each study site between 1983 and 1999. Twelve trials were established at the Duchesnay research forest (DUC, 46°57′N, 71°40′W), 24 trials were established at the Mousseau research forest (SVE, 46°35′N, 74°58′W), and the remaining trials established across the province (Fig. 1). Initially, each trial was composed of paired experimental units (EUs). One EU was harvested (2 ha) and the other was left as an untreated control (1 ha). EUs were split into 0.25 ha sections (50 × 50 m) to facilitate inventory. Given logistical and financial constraints, however, only one or two 0.25 ha sections were re-measured over time for saplings. Hence, within each EU we combined data from all sections to avoid artificially increasing our sample size. In general, trials were re-measured every ten years. The actual number of EUs used varied according to each objective based on the presence of saplings of each species and diameter class, and silvicultural treatment. Single-tree selection cutting (∼30% merchantable BA removal) was most frequently used. Selection cuts were aimed at reducing mortality losses, improving stand quality, and maintaining an uneven-aged structure on a rotation that ranged from 15 to 25 years.

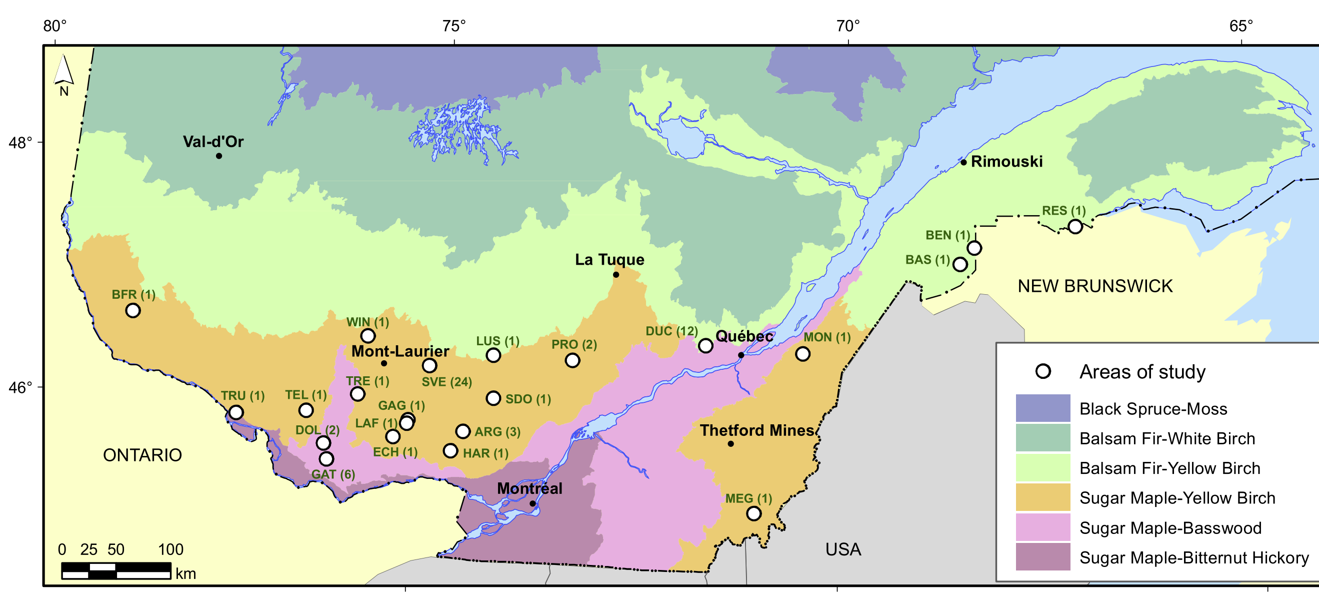

Fig. 1. Location of the 22 study sites in Québec, Canada. Sites are located within three bioclimatic domains (Saucier et al., 1998) along a latitudinal gradient: sugar maple – basswood, sugar maple – yellow birch, and balsam fir – yellow birch. Full names of study sites are: ARG, Argenteuil; BAS, Basley; BEN, Benedicte; BFR, Bois Franc; DOL, Lac Doley; DUC, Duchesnay; ECH, Lac Echo; GAG, Lac Gagnon; GAT, Gatineau; HAR, Harrington; LAF, Lac Lafontaine; LUS, Lusignan; MEG, Lac Mégantic; MON, Montmagny; PRO, Lac Provision; RES, Restigouche; SDO, Saint-Donat; SVE, Sainte-Véronique; TEL, Lac Telfer; TRE, Lac Trente-et-un-Miles; TRU, Lac-à-la-Truite; WIN, Lac Windigo. The number of cutting trials is shown in parentheses.

Table 1
Postharvest characteristics of mature northern hardwood stands from which saplings were sampled for objectives 1 and 2. Abbreviations: Trt, treatment; BA, basal area; EU, experimental unit; SE, standard error.

Trt	Sites	EU	Density (stems ha^{-1})		BA (all species)		BA (% maple)		BA (% birch)		BA (% beech)	
	n	*n*	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Cut	22	64	393	12	18.8	0.5	56	4	16	3	9	2
Uncut	22	51	510	18	26.2	0.5	53	4	19	3	10	2

Table 2
Preharvest and postharvest sapling density (mean ± SE) for the three most abundant sapling species among 22 study sites located in northern hardwood forests of Québec. Partial harvesting was generally carried out using single-tree selection cutting.

Species	Presence	Preharvest (saplings ha^{-1})	Postharvest (saplings ha^{-1})	Difference (saplings ha^{-1})	Difference (%)
Sugar maple	59	498 ± 53	333 ± 34	164 ± 24	31 ± 2
American beech	42	277 ± 54	203 ± 40	74 ± 17	19 ± 2
Yellow birch	48	34 ± 6	23 ± 4	11 ± 3	21 ± 3
Total	61	809 ± 69	559 ± 47	249 ± 29	30 ± 2

The presence of BBD has been documented in Québec. Currently, all sites west of DUC (Fig. 1) are in the advancing front stage (Shigo, 1972), where trees are infested with the scale insect, but not yet infested by *Nectria coccinea* (Pers.: Fr.) Fr. var. *faginata* Lohman, A.M. Watson, & Ayers or *Nectria galligena*, with low mortality due to BBD. Since 2009, the DUC, Montmagny (MON), and Lac Mégantic (MEG) sites are at the killing front stage, where *Nectria* infection is present and mortality is increasing. However, this mortality affects only a proportion of the largest beech trees in some EUs and only for the last five years of the last measurement period. The Basley (BAS), Benedicte (BEN), and Restigouche (RES) sites are in the aftermath stage, but BEN and RES sites have no overstory beech. Hence, the BAS site is the only study site where BBD has caused substantial beech mortality between measurements year 10 and 20 (1998–2008). Consequently, the effect of BBD on stand dynamics in our study is low.

2.3. Measurements

In each EU, all saplings were counted by species and DBH (2 cm, 1.1–3.0 cm; 4 cm, 3.1–5.0 cm; 6 cm, 5.1–7.0 cm; 8 cm, 7.1–9.0 cm). A conversion factor was used to express sapling density and BA as per-hectare values. Merchantable trees were also measured by species and DBH. Additional information was collected in each site: longitude (67.1–78.6°W), latitude (45.5–47.9°N), aspect (0–360°), slope (0–40°), drainage (rapid, well drained, moderate, and imperfect), and soil depth (20–101 cm). Mean annual temperature (1.7–4.6 °C) and precipitation (915–1390 mm) for the 1981–2010 period were also estimated using the BioSIM software (Régnière et al., 1995).

2.4. Soil samples

Soil samples were also taken between 2010 and 2014, 15–30 years postharvest, from about two-thirds of the EUs. At least two composite samples were collected near the EU's outside edge. For each soil sample, a 40 × 40 cm square was cut in the forest floor (L-F-H) to collect the H horizon. The first 10 cm of the B horizon was also collected. Soil from the B horizon was placed in a bucket where particles >2 mm were removed. All soil samples were then placed in plastic bags and kept at low temperatures (<12 °C) in the field using a cooler and then frozen prior to chemical analyses. Variables measured were soil organic matter (OM), humidity, total C and N, pH ($-\log[H^+]$), K, Ca, Mg, Mn, Al, Fe, Na, as well as effective soil acidity, cation exchange capacity (CEC), and base saturation (BS). A 5–10 g sample of collected soil was dried at 105 °C to determine soil humidity, and charred at 550 °C overnight to determine OM (CSSS, 1993). Another 300–500 mg sample of collected soil was burnt at 1350 °C. Total C was measured by infrared absorption and total N was measured by thermal conductivity of gases produced during combustion (LECO TruMac CN Elemental Analyzer, St Joseph, MI, USA). A 10 g soil sample was diluted in 20 mL of demineralized water and soil pH was determined after 30 min using a calibrated pH meter (Walsh, 1971; McKeague, 1978). Exchangeable cations (K, Ca, Mg, Mn, Al, Fe, Na) were extracted with an unbuffered NH_4Cl (1 M) solution and measured by plasma atomic emission spectrometry (ICAP 9000, Thermo Instruments, Franklin, MA, USA for 2010–11 samples; Optima 4300 or 8300, Perkin Elmer, Waltham, MA, USA for 2012–14 samples; Espiau and Peyronel, 1976; Rouiller et al., 1980; Jones, 1999). Effective soil acidity represents the sum of H^+ and Al while CEC represents the sum of K, Ca, Mg, Na, and soil acidity. BS represents the proportion of CEC as base cations. All soil analyses were carried out internally by the Organic and Inorganic Chemistry Laboratory of the Ministère des Forêts, de la Faune et des Parcs du Québec.

2.5. Statistical analyses

We used linear regression to establish relationships between sapling abundance and ingrowth (objective 1), and test the influence of silvicultural treatment at the EU level. For each species, i.e., sugar maple, American beech, and yellow birch, postharvest sapling density (stems ha^{-1}) and BA ($m^2\ ha^{-1}$) were plotted against the respective density or BA of tree ingrowth in year 10 and in year 20. We tested whether the partial cut and uncut control treatments shared the same intercept and slope. Because most regressions did not share the same slope, regressions were presented separately by treatment. EUs had to have at least one sapling and one recruit of the same species and diameter class to be included in analyses. We used Cook's distance (Weisberg, 2005) to identify potential extreme values in our dataset. While we did not find any measurement errors that would justify removing potential extreme values, we carried out the regression with and without these potential extreme values to quantify their influence on the robustness of each regression (Weisberg, 2005).

To determine factors that help explain sapling abundance (objective 2), Pearson correlation coefficients were used at the EU level in absolute (# ha^{-1}) and relative (% of total) terms. Postharvest sapling density and BA of each treatment/species combination

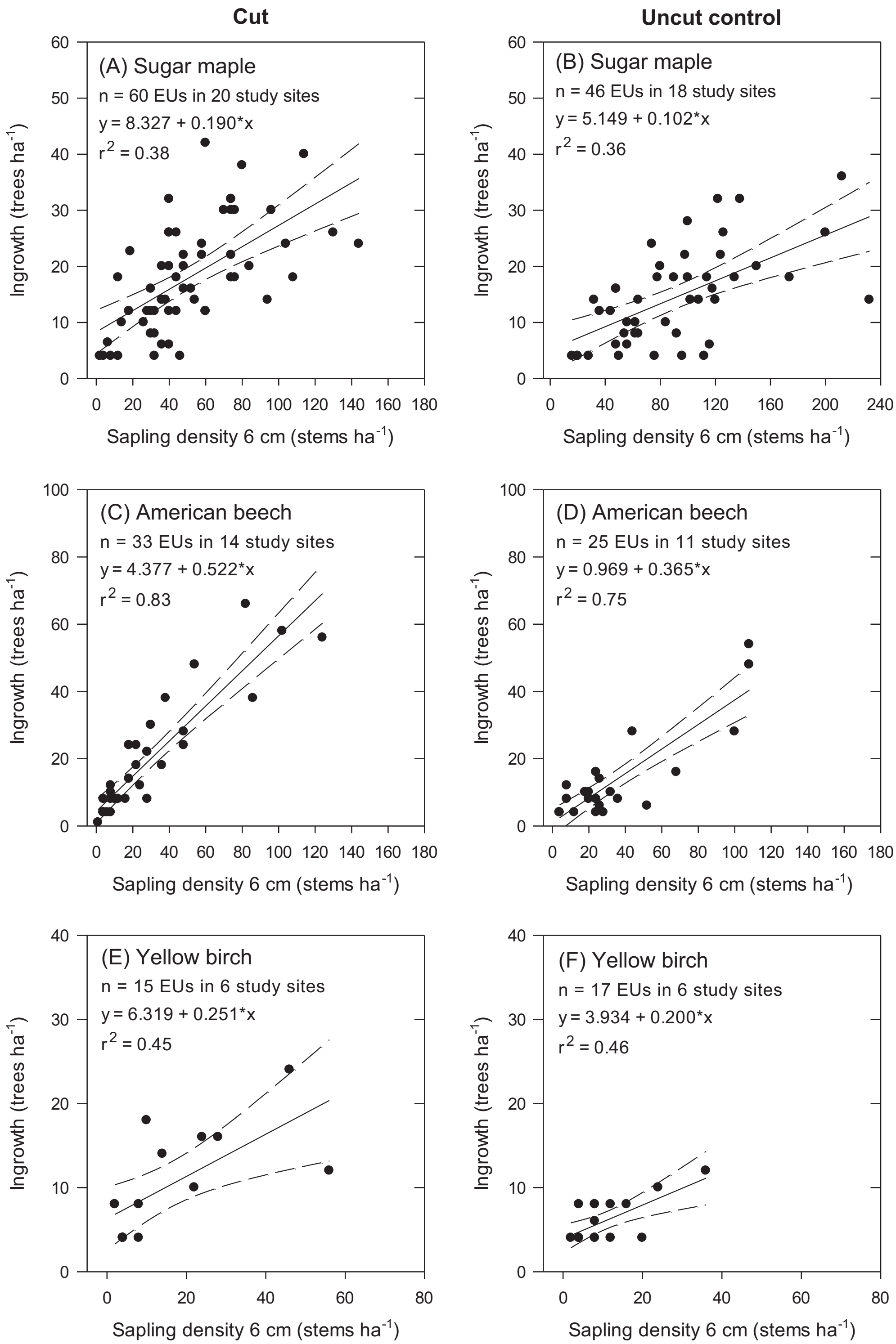

Fig. 2. Linear regression of sapling density ha^{-1} (stems 6 cm in diameter at breast height (DBH)) and ingrowth (trees ha^{-1}) after 10 years. Each point is an experimental unit, the solid line is the regression equation, and dashed lines are the 95% confidence interval. Regressions are presented separately for each species (sugar maple, American beech, yellow birch) and harvesting treatment (partial cut vs. uncut control).

were correlated against: postharvest tree density and BA of sugar maple, American beech, and yellow birch, longitude, latitude, aspect, slope, drainage, soil depth, mean annual temperature, and mean annual precipitation. Correlations were carried out with the following soil variables: OM, humidity, total C, total N, pH, K, Ca, Mg, Al, Fe, CEC, BS, effective soil acidity, as well as variables used to determine Ca, K, and P deficiencies in sugar maple (Ouimet et al., 2013): percent saturation in Ca, percent saturation in Mg, Ca Mg^{-1} ratio, and K Mg^{-1} ratio. Correlations with soil data were conducted using the sapling inventory date closest to the soil sampling date.

To quantify the change in sapling abundance (objective 3), sapling density and BA of each treatment/species combination were plotted over time and a random coefficient regression model was

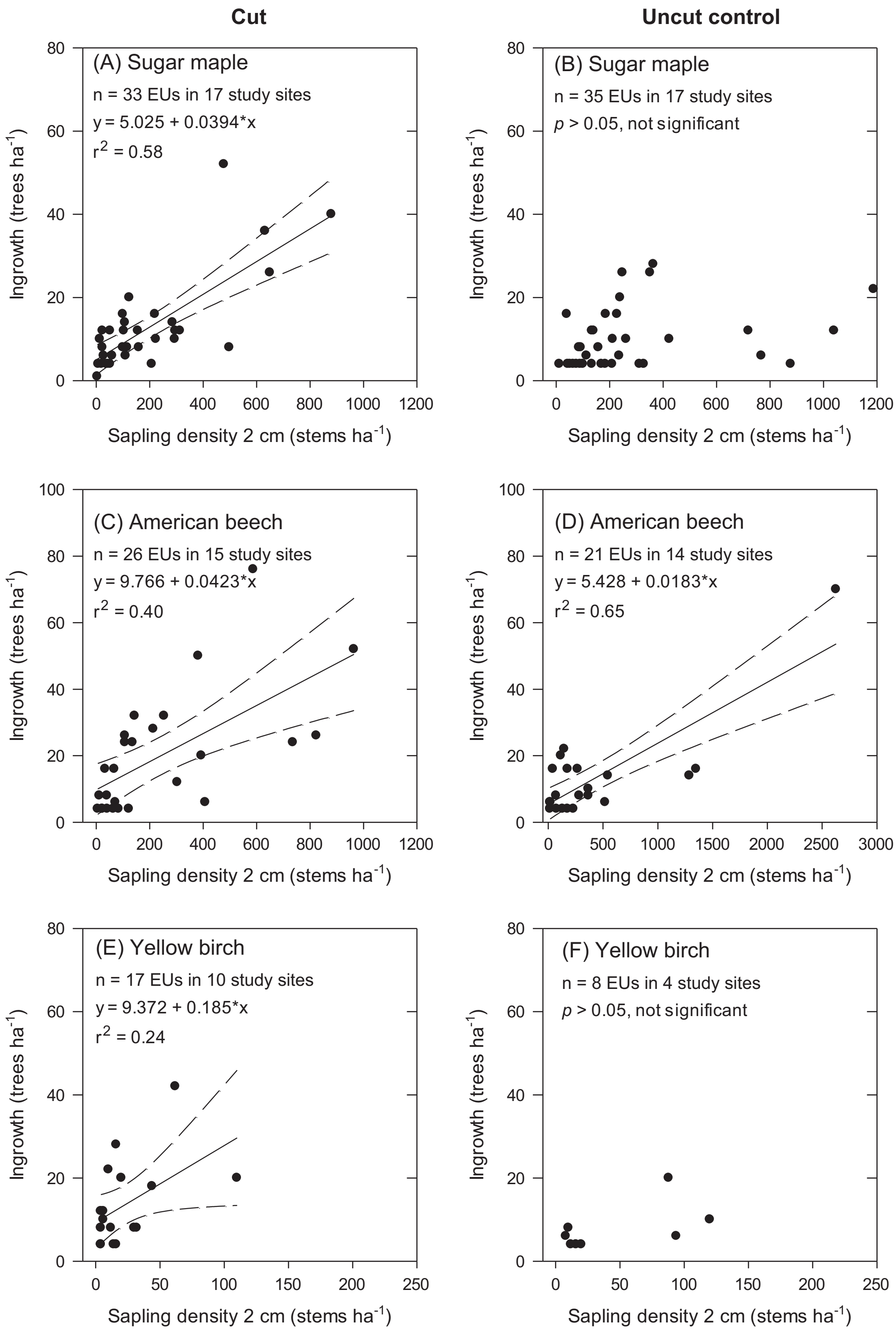

Fig. 3. Linear regression of sapling density ha^{-1} (stems 2 cm in diameter at breast height (DBH)) and ingrowth (trees ha^{-1}) after 20 years. Each point is an experimental unit, the solid line is the regression equation (when $p < 0.05$), and dashed lines are the 95% confidence interval. Regressions are presented separately for each species (sugar maple, American beech, yellow birch) and harvesting treatment (partial cut vs. uncut control).

used to test whether each slope was significantly different from zero. The EU was used as the subject of repeated measures. Such models are appropriate when repeated measurements do not occur at fixed intervals, or when the relationship with time is of particular interest (Brown and Prescott, 2006).

Each regression or correlation was carried out with SAS version 9.4 (SAS Institute Inc., Cary, NC). Statistical significance was defined as $p < 0.05$. Several combinations of sapling diameter classes (2, 4, 6, 8 cm) and variables (density, BA) were tested, but only those with the best fit, most frequent occurrence among the range of species and treatments, and highest management implications were presented in this article. Best fit criteria refers to highest r or r^2 (min $r^2 \sim 10\%$) for objectives 1 and 2, and lowest Akaike Information Criterion (AIC) for objective 3. Normality of residuals and homogeneity of variance postulates were verified. A squared root transformation was necessary for sapling BA (objective 3).

Table 3
Soil chemical properties of H and B horizons collected from 22 study sites in partially cut and uncut northern hardwood stands of Québec, Canada. Abbreviations: BS, base saturation; Eff. Acid., effective acidity; OM, organic matter; Ca sat, Calcium saturation; CEC, cation exchange capacity; SE, standard error; Silv Trt, silvicultural treatment.

Silv Trt	Variable	Units	H horizon				B horizon			
			Mean	SE	Min	Max	Mean	SE	Min	Max
Partial cut (n = 54)	OM	g kg^{-1}	661	25.5	197	931	109	4.8	61	212
	Humidity	%	7	0.4	3	20	4	0.2	2	11
	pH	(−log[H$^+$])	4.6	0.1	3.9	5.4	5.0	0.1	4.4	5.9
	C	g kg^{-1}	337	13.0	90	483	46	2.2	23	98
	N	g kg^{-1}	17	0.6	6	25	2	0.1	1	5
	K	mg kg^{-1}	641	35.0	152	1148	36	2.2	14	97
	Ca	mg kg^{-1}	3246	236.7	332	9383	164	22.5	22	806
	Ca sat	%	64	2.3	9	86	17	1.9	3	60
	Mg	mg kg^{-1}	327	19.0	66	649	16	1.4	5	55
	Mg sat	%	11	0.4	5	17	3	0.2	1	8
	Ca Mg^{-1}	Ratio	10	0.4	2	20	9	0.8	2	38
	K Mg^{-1}	Ratio	2	0.1	1	4	3	0.1	1	5
	Al	mg kg^{-1}	142	27.7	3	1132	297	19.4	89	771
	Fe	mg kg^{-1}	30	4.0	1	153	24	3.2	2	131
	Eff. Acid.	meq 100 g^{-1}	2	0.3	0	14	3	0.2	1	9
	CEC	meq 100 g^{-1}	23	1.2	8	55	4	0.2	2	10
	BS	%	83	2.6	20	100	23	2.0	6	67
Uncut control (n = 45)	OM	g kg^{-1}	749	24.9	224	939	116	5.8	59	248
	Humidity	%	8	0.4	3	15	4	0.4	2	18
	pH	(−log[H$^+$])	4.4	0.1	3.8	5.2	4.9	0.1	4.5	5.5
	C	g kg^{-1}	382	12.7	117	478	49	2.6	23	95
	N	g kg^{-1}	19	0.6	6	25	3	0.1	1	5
	K	mg kg^{-1}	700	41.7	152	1565	37	2.6	15	81
	Ca	mg kg^{-1}	3263	199.7	592	7475	126	17.2	30	700
	Ca sat	%	65	1.9	30	84	14	1.4	4	53
	Mg	mg kg^{-1}	354	20.9	67	718	16	1.5	4	54
	Mg sat	%	12	0.5	5	19	3	0.2	1	7
	Ca Mg^{-1}	Ratio	9	0.3	5	17	8	0.5	3	19
	K Mg^{-1}	Ratio	2	0.1	1	4	3	0.1	1	4
	Al	mg kg^{-1}	143	28.2	4	667	331	22.6	102	773
	Fe	mg kg^{-1}	30	6.1	2	252	29	4.6	2	159
	Eff. Acid.	meq 100 g^{-1}	3	0.4	0	9	4	0.3	1	9
	CEC	meq 100 g^{-1}	24	1.0	8	46	5	0.3	2	10
	BS	%	85	2.3	40	99	20	1.5	7	60

3. Results

3.1. Relationships between saplings and ingrowth after 10 and 20 years

In cut stands, the strongest linear relationships were found between sapling density in the 6 cm class and tree ingrowth after 10 years (Fig. 2A, C, and E). Percent variation explained was 38% for sugar maple, 83% in beech, and 45% in yellow birch. Similar relationships were found in control stands, with 36% in maple, 75% in beech and 46% in birch (Fig. 2B, D, and F). Linear relationships with density of other diameter classes (e.g., combination of 6 and 8 cm) or with sapling BA were also found, but they were not as strong as those with sapling density in the 6 cm class among all species and treatments (Appendix A and B).

After 20 years, linear relationships were found with sapling density in the 2 cm class and ingrowth in cut stands (Fig. 3A, C, and E). Percent variation was 58% for sugar maple, 40% in beech, and 24% in yellow birch. Relationships in control stands were inconsistent, with an r^2 value of 65% in beech and lack of significance in maple and birch (Fig. 3B, D, and F). The relationship in beech (Fig. 3D) is the only case where the removal of a potential extreme value caused the regression to become non-significant (Appendix A and C). Other diameter classes also had some influence on ingrowth (e.g., combination of 2 and 4 cm, 4 cm alone, see Appendix C and D), but the 2 cm class generally had the best fit among species and treatments.

For sugar maple and American beech, regression of partial cut and uncut control treatments shared the same intercept but not the same slope (Appendix A and C). The slope was higher after partial cutting. For yellow birch, both treatments shared the same intercept and slope.

3.2. Factors explaining sapling abundance

On average, soil data was collected 2 years after the closest sapling measurement (median = 0 years), and 95% of soil data was collected within 3 years of a sapling measurement. Sandy loam was the most abundant textural class (78%), followed by loam (9%), and loamy sand (5%). Study sites sampled were acidic (Table 3), and 88% of EUs had Ca deficiencies for sugar maple health ($\leqslant$28.4% in B horizon, Ouimet et al., 2013). Only 6% of EUs, however, had K deficiencies for sugar maple health (as indicated by Ca Mg^{-1} $\leqslant$ 4.382 in H horizon, Ouimet et al., 2013). Close to 22% and 62% of EUs had BS of the B horizon <12% and <20%, respectively. BS values <12% can limit sugar maple to <20% of total regeneration, while BS values <20% can limit sugar maple to <60% of total regeneration (Sullivan et al., 2013).

Correlations with sapling density and relative density were stronger than those with sapling BA. Postharvest density of saplings 6–8 cm DBH ha^{-1} was positively correlated to density of merchantable trees of each respective species in cut and uncut stands, with values ranging from 0.43 to 0.75 (Fig. 4). In uncut EUs, relative sugar maple density was positively correlated with saturation of BS and Ca in the B horizon (Fig. 5A and B) and negatively correlated with effective soil acidity in the H horizon (Fig. 5C). Similar correlations were also found in cut EUs, but they were not as strong (r = 0.2–0.3 range, data not shown). All other variables tested showed weak relationships (r < 0.4) with sapling abundance.

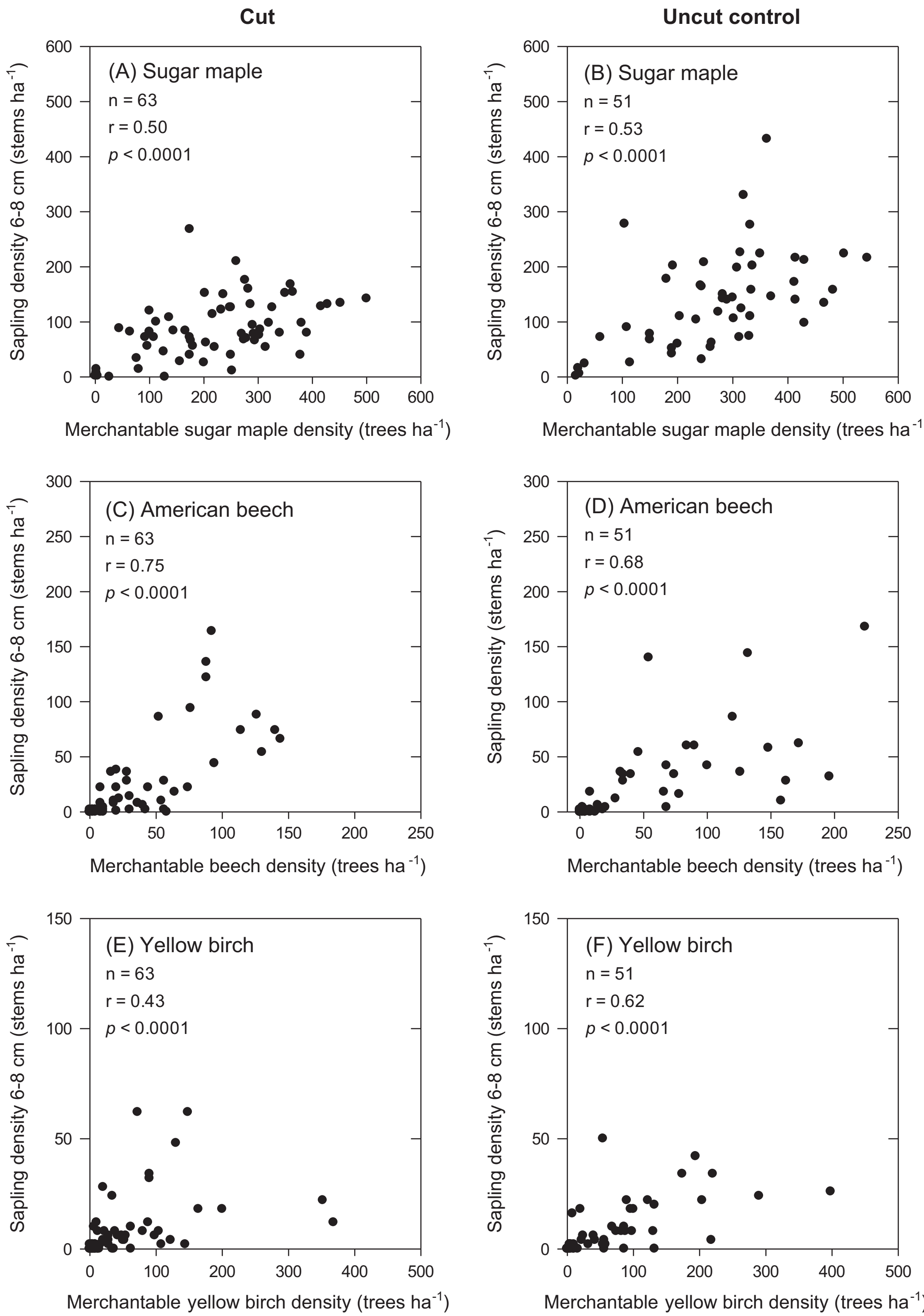

Fig. 4. Significant ($p < 0.05$) correlations between the postharvest number of saplings 6–8 cm in DBH and postharvest density of merchantable sugar maple (A and B), American beech (C and D), and yellow birch (E and F). Partially cut stands are shown on the left column and uncut control stands are shown on the right column.

3.3. Sapling change over time

In all cases, BA offered a better model fit (lower AIC) than density regardless of data transformation. In cut stands, sugar maple and yellow birch sapling BA showed slight increases of 0.18 and 0.23 $m^2\ ha^{-1}$ over the 30-year period, respectively (Fig. 6A and E). American beech BA doubled during the same period, from 0.55 to 1.33 $m^2\ ha^{-1}$ (Fig. 6C). In uncut control stands, BA of sugar maple declined slightly by 0.09 $m^2\ ha^{-1}$ (Fig. 6B), yellow birch sapling BA showed no significant change (Fig. 6F), while beech sapling BA doubled from 0.59 $m^2\ ha^{-1}$ to 1.20 $m^2\ ha^{-1}$ (Fig. 6D).

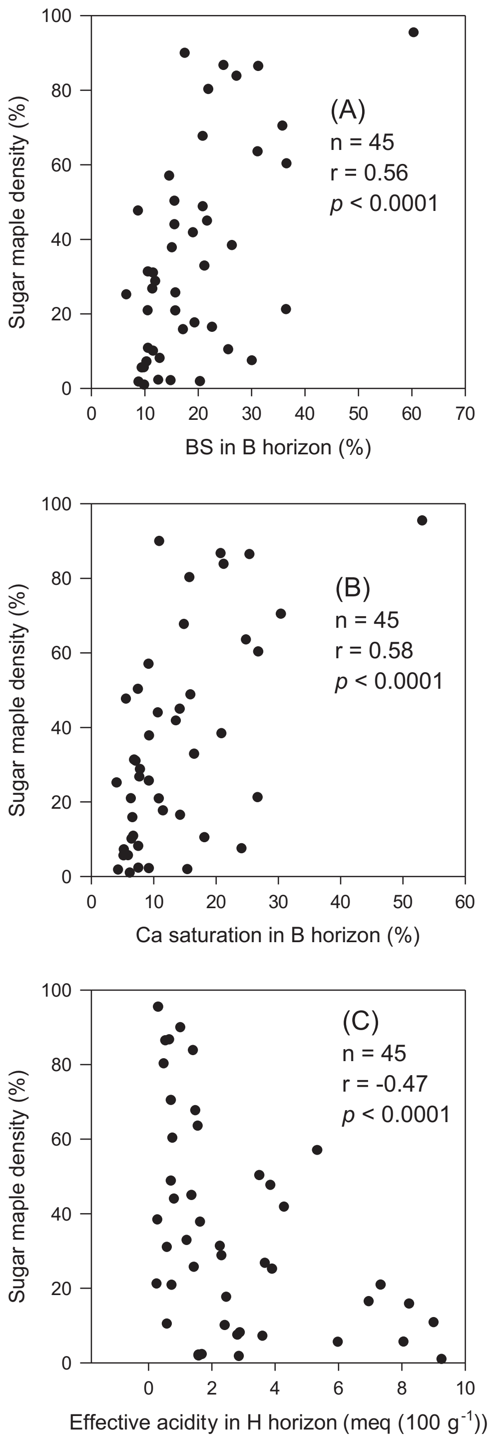

Fig. 5. Significant ($p < 0.05$) correlations between sapling density of sugar maple (% of total regeneration) and percent saturation of base cations in the B horizon (A), percent saturation of calcium in the B horizon (B), and effective acidity in the H horizon (C) in uncut EUs.

4. Discussion

4.1. Relationships between saplings and ingrowth after 10 and 20 years

In accordance with our first objective, we were able to establish linear relationships between sapling density and tree ingrowth for northern hardwood stands. Relationships varied among (1) species, (2) sapling size, (3) silvicultural treatment, and (4) time after harvest.

In terms of species differences, relationships for the first 10 years were generally strongest (higher r^2) in American beech, followed by yellow birch, and sugar maple. This indicates that larger saplings of each species can reach merchantable size within a decade, but in terms of growing space (Oliver and Larson, 1996), beech appears to be a better competitor. Lower sapling mortality in beech is one potential explanation. Because saplings were not tagged, however, we cannot exclude the possibility that some saplings remained alive but didn't grow enough to reach merchantability. Moore and Ouimet (2006) suggested that increased success of beech recruitment was related to depletion of base cations due to soil acidification. Deficiencies in base cations, such as Ca, are often correlated with sugar maple mortality as recently reviewed by Bal et al. (2015). After 20 years, sugar maple had the strongest relationship between saplings and ingrowth in cut stands, followed by beech and birch. The weaker relationship in yellow birch may be related to the small number of EUs that could be used in regression analyses.

In terms of sapling size, ten-year ingrowth was most strongly associated with saplings in the 6 cm diameter class. This is somewhat surprising, considering that the 8 cm diameter class should be more closely connected to short-term ingrowth. Given that occurrence, postharvest density, and injury from harvesting were similar between the 6 and 8 cm size classes, the explanation lies elsewhere. We suspect this may be related to the physiological age of saplings. If saplings in the 8 cm class spent more of their lifetime in a suppressed condition than saplings in the 6 cm class, it could have reduced their ability to respond to release and reach merchantability. Donoso et al. (2000) showed that saplings that grew for at least 80% of their life in a managed condition, such as after selection cutting, needed less time to reach a given diameter compared to saplings that spent less than 67% of their life in a managed condition. Hence, suppression might have an effect on crown architecture which reduces the sapling's capacity to respond to release.

After 20 years, the 2-cm diameter class became more closely associated with recruitment than larger sapling sizes. This suggests that smaller saplings should be taken into account for longer rotation periods. While we did not find any measurement error associated with the potential extreme value found in Fig. 3D, its substantial influence suggests that this particular regression equation should be used with caution. Other potential extreme values had less influence on r^2. Slopes of regression lines (Figs. 2 and 3) also provide some indication of the potential number of saplings needed to obtain at least one recruit. After 10 years in cut stands, six sugar maple saplings in the 6 cm class were needed to get one recruit ($6 \times 0.190 \geqslant 1$) compared to two beech saplings ($2 \times 0.522 \geqslant 1$) and four yellow birch saplings ($4 \times 0.251 \geqslant 1$).

The influence of silvicultural treatment on sapling-ingrowth relationships was also present for sugar maple and beech. For a given sapling density >0, tree ingrowth was higher in cut stands compared to uncut stands. Increases in growing space after partial harvest likely provided better growing conditions for saplings, such as increased light. This also increased their probability of survival until reaching merchantable size. As for yellow birch, the lack of influence of single-tree selection cutting may be related to low sample size and the fact that this intermediate shade tolerant species is rarely present as advanced regeneration and requires more severe disturbance to regenerate adequately.

Time after harvest also appeared to influence sapling-ingrowth relationships, with generally stronger relationships after 10 years than after 20 years (except sugar maple in cut stands). This can be explained by the lower number of replicates re-measured after

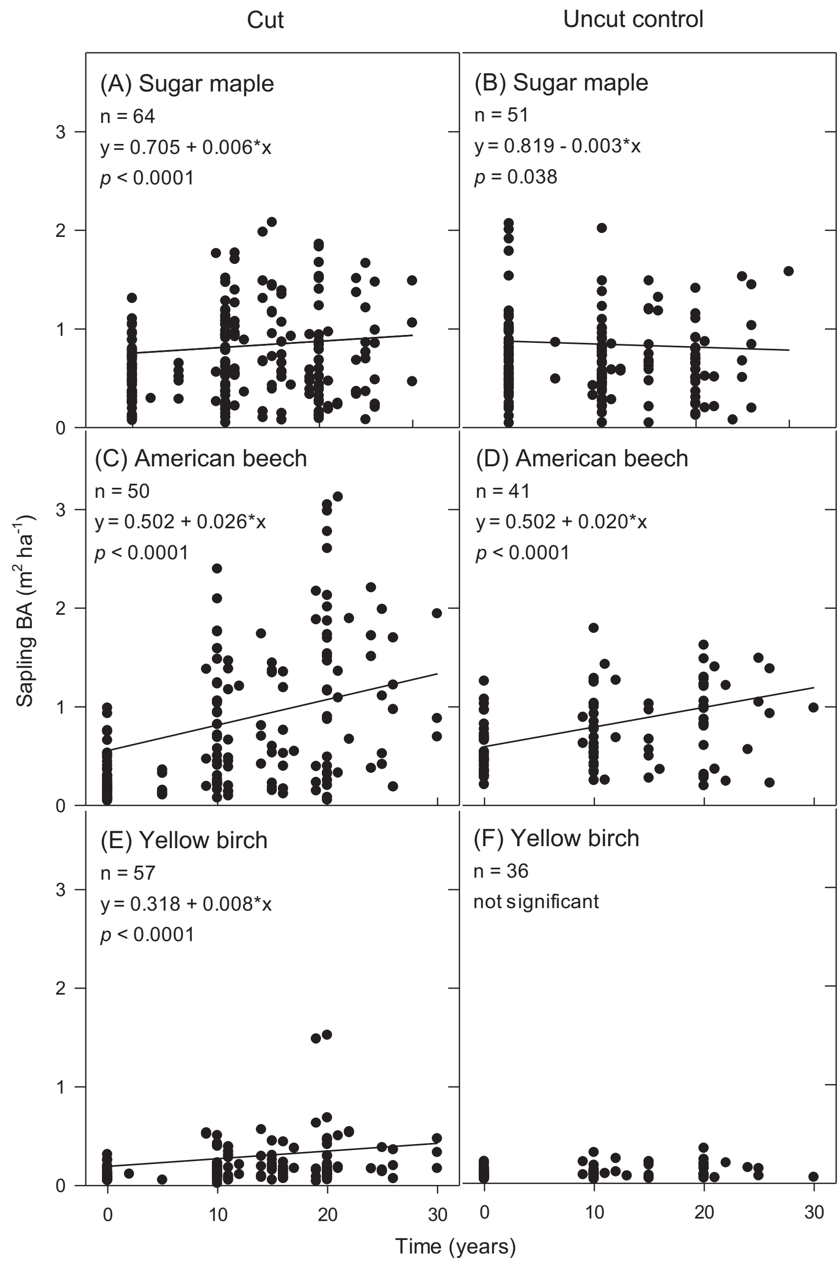

Fig. 6. Change in sugar maple (A and B), American beech (C and D), and yellow birch (E and F) sapling basal area (BA) over time based on data from 22 study sites across Québec, Canada. All sapling diameter classes were combined per species. Untransformed data points are shown. Regression line based on equation from transformed data (squared root).

20 years compared to after 10 years, and also because variation tends to increase over time.

4.2. Factors explaining sapling abundance

None of the variables tested showed a strong correlation (e.g. $r \geqslant 0.80$) with sapling abundance. This highlights the stochastic nature of the early phases of tree development, in this case the sapling stage. Variables that were most strongly correlated with sapling abundance were related to the density of mature trees in the overstory, regardless of silvicultural treatment. This result is corroborated by several studies (e.g., Canham et al., 2006; Béland and Chicoine, 2013; Graignic et al., 2013). Béland and Chicoine (2013) found that sugar maple sapling density was positively related to the proportion of mature sugar maple in the residual stand. Relationship strength varied from 0.34 to 0.38 depending upon sapling size. Graignic et al. (2013) reported that sugar maple seedling density increased with increasing basal area of mature sugar maple in uncut stands. Canham et al. (2006) also found that the immediate configuration of the residual stand after partial cutting showed a strong influence on long-term yield in New England forests. Our correlations may be explained by stand structure. Stands that have higher tree density in pole-size classes (10–24 cm) are more likely to have a larger number of saplings.

As for soil properties, positive correlations between relative sugar maple density and BS and Ca saturation are consistent with results from the literature (Ouimet et al., 2013; Sullivan et al., 2013). Effective soil acidity can serve as a proxy for the inverse of Ca saturation, hence the negative correlation with sugar maple density is also consistent with the literature. Correlations with soil properties were stronger in uncut stands compared to cut stands, particularly for BS and Ca saturation. We think this is related to variability among treatments; standard deviation was 4% higher in BS and Ca saturation of cut stands compared to uncut stands. Higher intra-treatment variation is more likely to result in lower correlation values.

Otherwise, none of the correlations with soil properties were very strong. Bannon et al. (2015) found a lack of correlation between soil nutrients and density of sugar maple and American beech saplings based on data from 56 sugar maple stands in southern Québec. Our results showed that most EUs were located on acidic sites with low Ca saturation. We hypothesize that the range of soil conditions was not large enough to establish stronger correlations. Other potential explanations include total nutrient status as the only form of soil data, and the delay between soil data collection and sapling measurement. Lack of correlation between sapling abundance and climate/site physiography suggests that variables at a smaller scale (light and water, e.g., Hane, 2003; Kobe, 2006) or spatially-explicit competition indices (Hanson et al., 2011) may also be important in explaining sapling dynamics than variables at the stand scale.

4.3. Change in sapling basal area over time

In cut EUs, the increase in American beech sapling BA over the 30-year period was substantial compared to smaller increases found in sugar maple and yellow birch. This suggests that once established, beech saplings are better competitors for growing space in the understory on these study sites. In uncut control EUs, the increase in beech sapling BA was just as pronounced. Given the lack of increase in sugar maple and yellow birch, beech is likely more competitive than these two species even under low light environments.

Underlying mechanisms of increased competitiveness in beech can be attributed to a number of factors in the literature. The cation depletion hypothesis is prominent (e.g. Moore et al., 2012; Marlow and Peart, 2014). Gravel et al. (2011) also documented a relative increase in beech saplings at the expense of sugar maple saplings in maple-beech stands of southern Québec. Given similar growth rates between the two species, however, authors suggested these results could not be explained by the base cation hypothesis alone. Preferential browsing by increasing white-tailed deer populations on maple (Long et al., 2007; Matonis et al., 2011) may be another contributing factor, along with asexual reproduction in beech (Jones and Raynal, 1986; Jones et al., 1989; Beaudet and Messier, 2008; Wagner et al., 2010), or intense shading by beech regeneration (Hane, 2003). Compared to seedlings, root suckers have greater height and diameter growth, as well as lower mortality rates (Beaudet and Messier, 2008). Additionally, from a sexual reproduction standpoint, sugar maple and yellow birch saplings have a shallow root system, making them less competitive than American beech which develops a taproot (Burns and Honkala, 1990).

5. Management implications

Our results highlight several management implications for northern hardwood forests similar to the ones studied, i.e., dominated by sugar maple, with acidic soils depleted in base cations. It appears worthwhile to collect information such as sapling species composition and size during forest inventory because tree ingrowth varies accordingly depending upon rotation period and silvicultural treatment. Sapling abundance was most strongly correlated with species composition of the mature overstory, and to a lesser extent, soil BS and Ca saturation. Hence, soil data may not be as critical unless strong BS and Ca deficiencies are suspected. Potential sapling densities required to obtain at least one recruit are estimated depending upon the aforementioned factors.

Sapling data can also be used to help determine if there are enough saplings to ensure sufficient recruitment and long-term sustainability. Morneault et al. (2010) carried out a similar exercise using a diameter distribution for sawlog production for a hardwood forest located in Ontario, Canada. By extrapolating the diameter distribution into smaller size classes (2–8 cm DBH), a target of 380 saplings ha^{-1} (all species combined) may be needed after harvest to maintain a reverse *j*-diameter distribution with a *q*-value of 1.18. Thus, mean postharvest sapling density in our study (559 saplings ha^{-1}) should be enough to ensure sufficient ingrowth, with 62% of all cut EUs above the threshold. Still, this leaves a rather large portion of EUs (38%) below the minimum value of 380 saplings ha^{-1}. If only desired tree species are considered, close to 66% of cut EUs did not have at least 380 saplings ha^{-1} of sugar maple and yellow birch. This is exacerbated by the small sapling increase of sugar maple and birch measured over 30 years in cut EUs compared to the large increase in beech.

Given the current mortality of American beech due to BBD, few beech saplings will make it to sawlog size but they appear to hinder sugar maple and yellow birch ingrowth. To promote sugar maple in areas where herbicide use is not permitted (such as Québec), silvicultural options include cutting the understory beech prior to partial overstory removal. Soil scarification in gap openings would benefit yellow birch and sugar maple establishment (Bédard and DeBlois, 2010). Some studies suggest that increasing the severity of disturbance could help favor sugar maple over beech (Nolet et al., 2008), while other studies indicate such treatments may not be sufficient to favor sugar maple (Bannon et al., 2015). Liming can help promote sugar maple regeneration where soils are acidic and severely depleted in base cations (Moore et al., 2012), although beneficial effects can be influenced by several factors (Cleavitt et al., 2011).

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.foreco.2015.09.020.

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